

Harshith Suresh

San Jose, CA | 408-549-4216 | suresh72@purdue.edu
www.linkedin.com/in/harshith-suresh-056079216/ | github.com/Harshith24

EDUCATION

Purdue University

B.S. in Computer Science

May 2026
West Lafayette, IN

- Dean's List and Semester Honors, **AWS Solutions Architect Associate**
- Relevant Courses: Data Structures, Algorithms, Cybersecurity and Networks, Operating Systems
- Minors: Applications In Data Science, Entrepreneurship and Innovation

PROFESSIONAL EXPERIENCE

Software Engineering Intern

Tesla Inc.

August 2025 – December 2025

Fremont, CA

- Built a real-time **network topology** and traceroute simulations to visualize physical device connections and metrics, packet flow, and alerts, helping teams identify congestion, and debug network and hardware issues **twice as fast**.
- Architected a **text-to-query** pipeline with **LangGraph** and a fine-tuned **SLM** through **LoRA** to generate queries for **ClickHouse** and **PostgreSQL** databases, enhancing real-time data visibility and accessibility across our network.
- Developed a network audit tool to validate **100K+** network devices, reducing audit time from **hours to seconds**.
- Enhanced our network scraper to collect routing tables, vlans, port-channels, and **BGP** peers, from global Tesla network.
- Performed network automation tasks on multiple vendors such as password rotation, config updates, and **ZTP**.

Cybersecurity Consultant Intern

Crowe LLP

June 2025 – August 2025

Chicago, IL

- Performed cybersecurity assessments and **HITRUST** compliance reviews with public sector clients including technical configuration reviews including Active Directory, Entra ID, network architecture, and incident management.
- Conducted **penetration testing** assessments and leveraged tools such as Nmap, Nessus, Metasploit, and Wireshark.
- Automated deployment of Kali Linux and Windows VMs using **Ansible**, **Packer**, and **Bash** scripting.

Software Engineering Intern

Mechasense.io

December 2024 – April 2025

Remote

- Designed a script on AWS to ensure high availability, scalability, and security of our application by automating deployment through **Jenkins**, OS and software upgrades, and minimizing downtime at no additional costs.
- Researched and Implemented machine learning models including **KNN**, **XGBoost**, and **neural networks**; Achieved **90%** accuracy on neural network to detect and classify HVAC failures.
- Built a synthetic data generation script to supplement real-world datasets, enabling robust model training and testing in the absence of sufficient raw data

Security Engineering Intern

Munich Re

June 2024 - August 2024

New York, NY

- Developed a **Malware Scanning** pipeline on **Azure** with **Defender for Cloud** to isolate malicious files from our **SFTP** server through **NFS** and **REST API** and logging data to **Datadog** for auditing and investigation.
- Configured **IAM policies**, security group rules, and **subnets** to prevent unauthorized network and data access.
- Deployed infrastructure such as private endpoints, event handling, storage accounts and alerts through **Terraform**.
- Contributed to the Developer Community by addressing gaps between Azure and Terraform documentation.

Security Engineering Intern

SecureX.ai

June 2022 - August 2022

San Jose, CA

- Built a **BeautifulSoup** based web-scraper to extract **CVEs**, **CVSS** scores, and exploit data from **NVD** and **ExploitDB**.
- Contextualized threats based on environment and asset criticality to give us the true threat impact beyond CVSS scores.
- Containerized microservices using **Docker** and deployed them to **EKS cluster** through **Jenkins** pipeline.

PROJECTS

Lecture Study Assistant

Personal Project

December 2025

San Jose, CA

- Local **Video RAG** system for analyzing lecture videos and creating quizzes and concept reviews to bolster productivity.
- Integrated **EasyOCR** for image text extraction and **Whisper** for audio transcription and **PGvector** to store embeddings.
- Employed **Mistral** LLM and **Nomic** Embedding model and integrated with **FastAPI** and **Next.js**.

Library Management Project

Personal Project

December 2024

West Lafayette, IN

- Developed library management web app with Python, Flask, React, and SQLAlchemy
- Deployed on **GCP** App Engine with **MySQL** instance, using **Nginx** and **Gunicorn** for scalable request handling
- Optimized app with **normalization**, **indexes**, and **serializable concurrency** to ensure conflict free borrowing

SKILLS

- Python, Terraform, AWS, Azure, Java, Typescript, C, REST, SQL, PromQL, RDBMS, Mongo DB, Neo4J, Docker, RAG, LangChain, Linux, CI/CD, Network Protocols, Git, GCP, Network Audit Tools, Next.js, React